

Our Pipeline, Where It Fails, and Why Only LoRA Fixes It

1. How our current pipeline works

We turn a manuscript into a finished book in a chain of steps. Two of them use AI image models; the rest are ordinary, reliable text/layout steps.

Manuscript (.docx)

■			
▼	PARSE	read the story, design each character, plan every page	(text)
▼	REFERENCES	draw a reference sheet for each character	(AI image)
▼	GENERATE	draw each page from the text	<< Text-to-Image model >>
▼	CORRECT	check each page vs the reference, fix	<< Image-to-Image model >>
▼	COMPOSE	place the caption text on the art	(code)
▼	BOOK	assemble the PDF	(code)

The two AI image steps are the only places consistency can go wrong:

- **GENERATE = Text-to-Image.** Draws a whole page from a written description (“two dogs at a stadium...”), guided by the reference sheet.
- **CORRECT = Image-to-Image.** Looks at the drawn page, compares each character to its reference, and edits the page to fix mistakes.

2. Where the Text-to-Image model fails

The generator draws **every page from scratch**. It reads the character description and the reference sheet as *inspiration*, not as an exact spec — so it re-invents the character a little each time.

- Coat colour and shade drift between pages.
- The cap or bandana appears, disappears, or changes.
- The character’s **size changes** page to page.
- The face and proportions wander.

This is not a quality problem — a fancier model draws *prettier* pages that are **just as inconsistent**. It is how the tool works: freehand, every time.

3. Where the Image-to-Image model fails

The corrector’s job is to *rescue* the drift after the fact. It helps, but it cannot reach perfection, for four reasons:

- **It only has a flat reference sheet**, not “this exact character in this pose.” So it can match colours and items, but not lock a face or a body it was never given for that angle.

- **Editing can't re-impose an identity the model doesn't hold.** To fix a wrong face it must partly redraw it — freehand again — so it drifts while fixing.
- **It misses subtle problems.** A slightly wrong ear, a small size change, a soft face — the checker often passes these, so they never get fixed.
- **Fixing one thing can break another.** Correcting a colour can nudge a pose; correcting a pose can soften a face. The result oscillates instead of settling.

In short: the corrector is a **patch on a leak**, not a fix for the pipe.

4. “Can we just add another Image-to-Image pass on top?”

It feels like stacking more correction passes should keep improving the page. It does not — and here is the honest reason:

- **Every pass is also freehand.** Each edit is another random draw, so it can undo the last fix as easily as improve it. We watched it flip a detail back and forth between passes.
- **You can't edit your way to a character the model never learned.** No number of “make it match the sheet” edits gives the model true knowledge of the face; it is always guessing from a flat picture.
- **Each pass costs time and money** and adds new risk (a good page can get worse).

So more image-to-image = **more patches on the same leak**. It plateaus around “pretty good,” never “perfect,” and sometimes goes backwards.

5. Why only LoRA actually solves it

Everything above is a *cure-after-the-fact*. LoRA fixes the **cause**.

A **LoRA** teaches the image model the character itself — we show it ~10–20 pictures of each character, and it **learns their identity into its own memory**. After that:

- **The Text-to-Image step draws the character correctly the *first* time** — same face, coat, cap and size on every page — because the model now *knows* them, not just reads about them.
- **We barely need the corrector at all**, because there is little drift to fix.
- **Scenes still look natural** — the model paints the learned character freshly into each new pose, angle and setting.

Text-to-Image	+ Image-to-Image	+ more I2I passes	LoRA
Same character every page	Closer, not exact	Plateaus	Yes

Natural, varied scenes	Yes	Yes	Yes
Fixes cause or symptom?	symptom	symptom	cause
Converges to "perfect"?	No	No	Yes (≈95%+)

The difference in one line: text-to-image and image-to-image *describe* or *patch* the character; **LoRA makes the model *know* the character** — so it is right from the start instead of corrected afterwards. That is why it is the industry-standard way to keep a character identical across many AI images, and the only method that gives us **consistent characters *and* natural pictures** at once.

What it takes

A small pay-as-you-go **fal.ai** (or Replicate) account and one API key — about **\$10 one-time** to teach all five characters, then pennies per page. Nearly everything else in our pipeline stays exactly as it is.